Assignment - 01

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Submitted to -
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Q1 Why DBMS is required and what is it??
A Database management system is software used to store, organise, manage and retrieve data efficiently. It is required to reduce data redundancy, maintain data consistency, provide security and supports backup & recovery.

Q2. Why is DBMS better than Traditional file system?
DBMS reduces duplication of data, avoids inconsistency provides security, supports multiple users at the same time which file system doesn't.

Q3. What is data Abstraction? Explain its levels.
Data abstraction hides internal storage details from users.

Three levels are:

- physical level - How data is stored
- Logical level - structure of database
- View level - User-specific data view

Q4 What is ~~data independence~~? the three-schema architecture? Why it's needed?

It divides database into three layers.

Users ⟶ External level (view) ⟶ Conceptual level (logical structure)
↓
Internal Level (physical storage)

Importance:-

- Provides data independence
- Improves security
- simplifies database design

Q5 What is data independence?

~~Q6 What is an Entity relationship (ER) Model?~~

It is ability to modify schema at one level without affecting another.

Types:

1 physical Data Independence
2 Logical Data Independence

Q6 What is an Entity relationship (ER) model??
A conceptual model used to represent:

- Entities
- Attributes
- Relationship.

Q7) What are integrity constraints? Why are they important in DBMS?
Integrity constraints are rules that maintain accessing & consistency of data

Types: Primary key, foreign key etc.

Importances:-

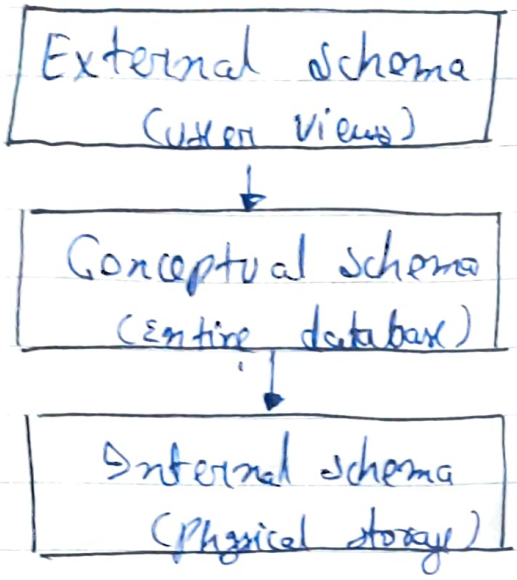
1) Maintain data Accuracy

2) Ensures data consistency

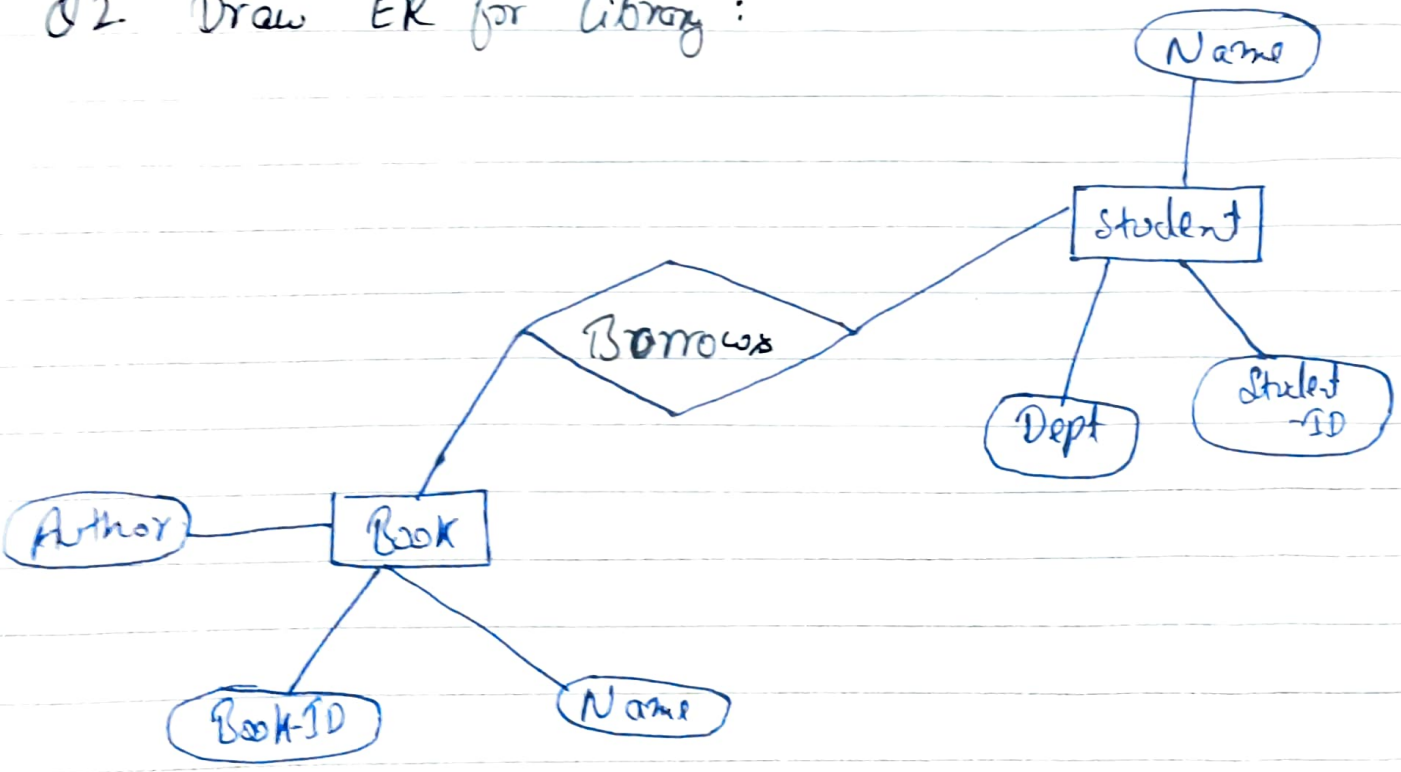
3) Avoids Invalid operations.

Section - B:

Q1 Draw the three schema level & explain.

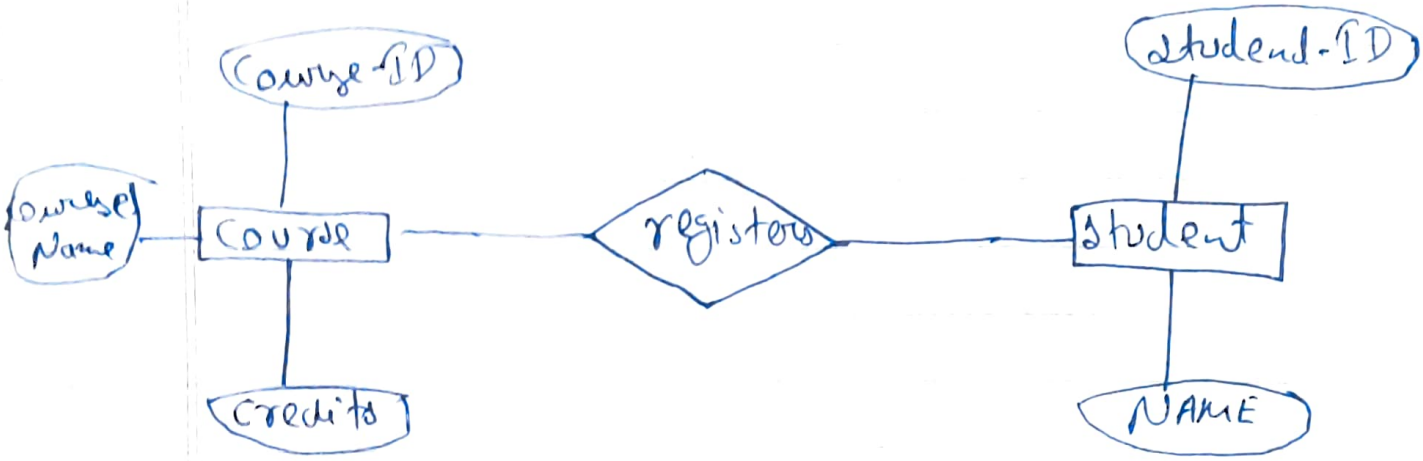


Q2 Draw ER for Library:



Q 3. Draw ER diagram for a student-course Registration system.

Many students ⟷ Many courses (M:N)



Section - C:

Q 1 A hospital wants to store patient, doctor & appointment data. Identify entities, attributes & relations.

Entities	Attributes
·) Patient	·) Patient (ID, Name, Age)
·) Doctor	·) Doctor (ID, specialization)
· Appointment	·) Appointment (Date, Time)

Patient → books → Appointment → with Doctor.

Q2. Online shopping

- Customer (customer_ID - primary key)
  Product (Product_ID - Primary key)
  Order (Order_ID - Primary key)

Constraints:

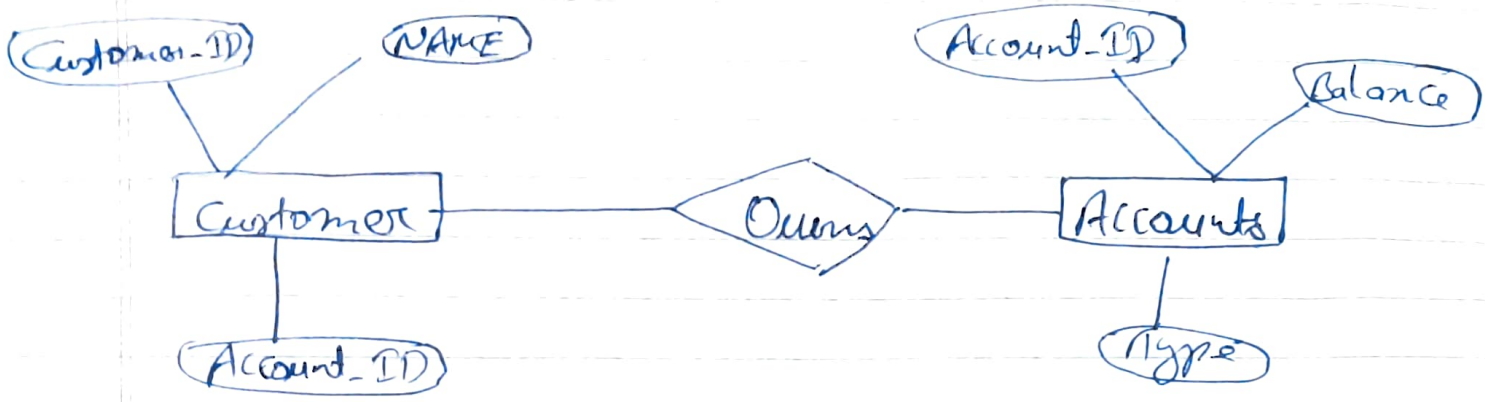
Primary key
not null
check (quality > 0)

Q3. A university stores student records. Explain how data abstraction helps.

Student see marks (view level)
Admin manages records (logical level)
DBA controls storage (physical level)

Section-D

Q1 Draw a ER diagram for any Real-life system.



Account_ID is a primary key
Relation: Customers Owns Account

Q2 Draw an Entity with attributes and Identify its Primary Key.

: Employee

- Employee_ID
- Name
- Salary
- Department

Employee_ID is primary key, because for all employee their ID will be different